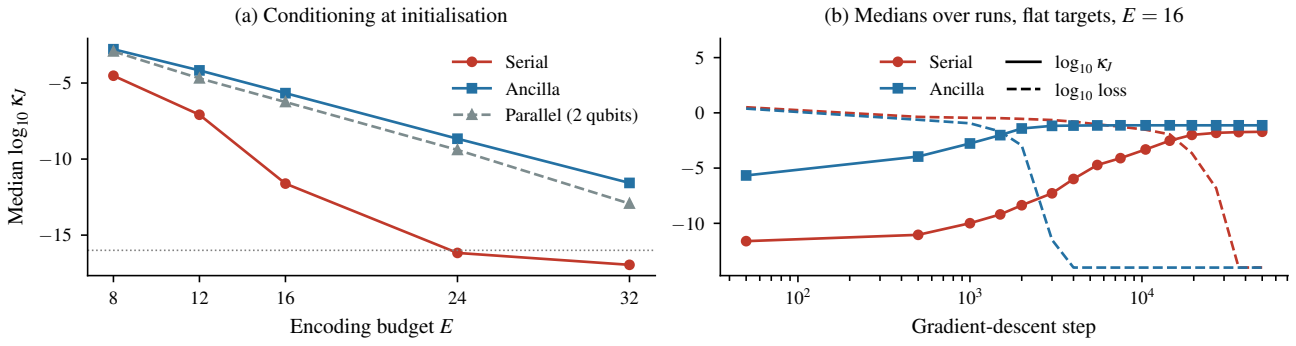

(a) Conditioning at initialisation
Median $\log_{10} \kappa_J$
−5
−10
−15
Serial
Ancilla
Parallel (2 qubits)
8
12
16
24
32
Encoding budget E
(b) Medians over runs, flat targets, $E = 16$
5
0
−5
−10
Serial
Ancilla
$\log_{10} \kappa_J$
$\log_{10}$ loss
10^2
10^3
10^4
Gradient-descent step